Joyride
Problem ID: joyride

It is another wonderful sunny day in July – and you decided to spend your day together with your little daughter Joy. Since she really likes the fairy-park in the next town, you decided to go their for the day. Your wife (unfortunately she has to work) agreed to drive you to the park and pick you up again. Alas, she is very picky about being on time, so she the told you when she will be at the park's front door and you have to be there at that time. You clearly also don't want to wait outside, since this would make your little daughter sad – she could have spend more time in the park!

Now you have to plan your stay at the park. Your know when you will arrive and when you will have to depart. The park consists of several rides, interconnected by small pavements. The entry into the park is free, but you have to pay for every use of every ride in the park. Since it is Joy's favourite park, you know how long using each ride takes and how much each ride costs. When walking through the park, you obviously must not skip a ride when walking along it (even if you have already used it), or else Joy would be very sad. Since Joy likes the park very much, she will gladly use rides more than once. Walking between two rides takes a given amount of time.

Since you are a provident parent you want to spend as little as possible when being at the park. Can you compute how much is absolutely necessary?

Input

The input starts with a number X, $1 \leq X \leq 1000$, the time between your arrival and the time you will be picked up (in minutes). The second line contains a three integers N, M and T, $1 \leq N, M, T \leq 1000$, the number of rides in the park, the number of pavements, and the number of minutes needed to pass over a pavement from one ride to another. You always start at ride 0 and have to return to ride 0 at the end of your stay, since the entry is located there. This means, that you have to use the ride 0 at least twice (once upon entry and once upon exit). It is allowed to take a ride more than once, if you have arrived at it.

The following M lines each contain two integers a and b, $0 \leq a, b < N$, stating that there is a pavement between the rides a and b. The following N lines each contain two integers t and p, $1 \leq t, p \leq N$, stating that the corresponding ride takes t minutes and costs p Euro.

Output

One integer, the minimum amount necessary to stay X minutes in the park, or if not possible `It is a trap`.

Sample Input 1	Sample Output 1
4 4 4 1 0 1 1 2 2 3 3 0 1 2 2 1 5 4 3 3	8

Sample Input 2	Sample Output 2
6 4 4 1 0 1 1 2 2 3 3 0 1 2 2 1 5 4 3 3	5